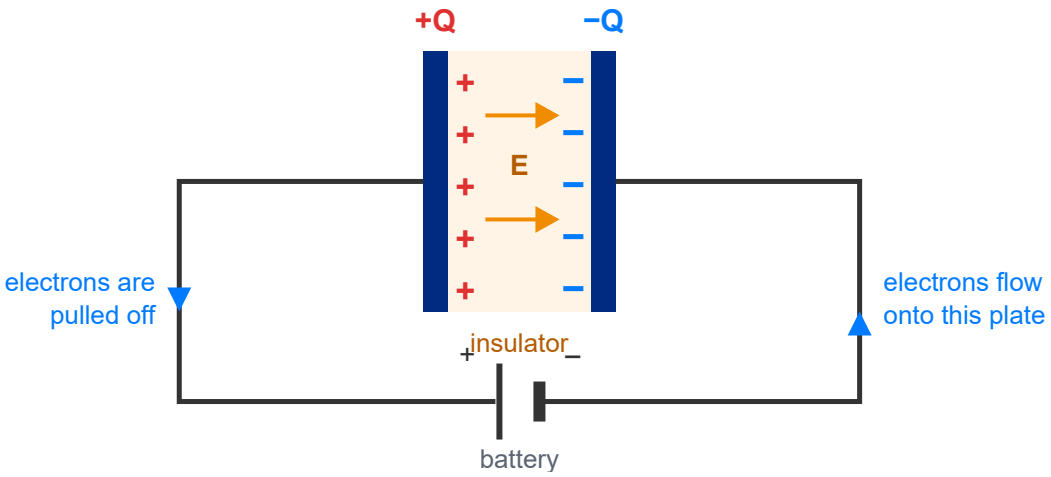
A2 · Unit 4 · WPH14 | Further Mechanics, Fields & Particles

Capacitors

A capacitor stores energy by holding positive and negative charge apart. It stores far less energy than a battery, but it can hand all of it back in a fraction of a second, which is why camera flashes, defibrillators and the smoothing circuit in every phone charger use one. On this page you will see where the ½ in $W = \frac{1}{2}QV$ comes from, why capacitors charge and discharge exponentially, and how a log graph turns that curve into a straight line.

What a capacitor is

The simplest capacitor is two parallel metal plates with an **insulator** between them (air, paper, plastic or a thin oxide layer). Connect it to a battery and the battery pushes electrons onto one plate and pulls the same number of electrons off the other. No charge ever crosses the gap. One plate ends up with charge $+Q$, the other with $-Q$.



Charging: the battery moves electrons round the circuit from the left plate to the right plate. The plates end up with equal and opposite charges, and an electric field across the gap.

As charge builds up, the plates develop a p.d. that **opposes** the battery. New electrons arriving at the negative plate are repelled by the ones already there. Charging stops when the p.d. across the capacitor equals the e.m.f. of the battery. At that point there is no p.d. left to drive a current, so the current is zero even though the circuit is complete.

COMMON MISTAKE

A capacitor does not store **net** charge: $+Q$ on one plate and $-Q$ on the other add up to zero. What it stores is **energy**, by keeping the charges separated. When a question says "the charge stored is Q" it means the charge on **one** plate.

Capacitance: C = Q/V

Push twice as much charge onto the plates and the field between them doubles, so the p.d. doubles too. Charge is **proportional** to p.d., and the constant of proportionality tells you how much charge the capacitor holds for each volt. That is its **capacitance**.

Capacitance

FORMULA SHEET

$$C = \frac{Q}{V}$$

C: capacitance / F Q: charge on one plate / C V: p.d. across the capacitor / V

DEFINITION

Capacitance is the charge stored per unit potential difference across the capacitor, $C = Q/V$. The unit is the **farad**: a capacitor has a capacitance of 1 F if it stores 1 C of charge when the p.d. across it is 1 V ($1\ \text{F} = 1\ \text{C V}^{-1}$).

One farad is huge. Real capacitors are usually measured in microfarads ($1\ \mu\text{F} = 10^{-6}\ \text{F}$), nanofarads ($10^{-9}\ \text{F}$) or picofarads ($10^{-12}\ \text{F}$). A 470 μF capacitor charged to 9.0 V holds $Q = CV = 470 \times 10^{-6} \times 9.0 = 4.2 \times 10^{-3}\ \text{C}$, a few millicoulombs.

Because $Q \propto V$, a graph of p.d. against charge is a straight line through the origin with gradient $V/Q = 1/C$. A bigger capacitor gives a **shallower** line: it takes more charge to raise its p.d. by each volt.

Energy stored in a capacitor

Charging a capacitor takes work. The first bit of charge goes on easily, because the plates are uncharged and the p.d. is zero. Every later bit has to be pushed onto a plate that already repels it, against a p.d. that keeps growing. The last bit is pushed against the full p.d. V.

Moving a small charge ΔQ through a p.d. V takes work $\Delta W = V\,\Delta Q$. On a graph of V against Q, that is the area of a thin strip of height V and width ΔQ. Add up all the strips and you get the **area under the graph**, which is a triangle.

Deriving W = ½QV from the area under the V–Q graph

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. To move a small extra charge ΔQ onto the plates when the p.d. is V, you do work (p.d. is work done per unit charge):

$$\Delta W = V\,\Delta Q$$

If ΔQ is small, V hardly changes while you add it.

2. On a graph of V (vertical) against Q (horizontal), $V\,\Delta Q$ is the area of a thin strip under the line. The total work done charging from 0 to Q is the sum of all the strips, which is the area under the graph:

$$W = \sum V\,\Delta Q = \text{area under the } V\text{–}Q \text{ graph}$$

3. Because $V \propto Q$, the graph is a straight line through the origin, so the area is a triangle with base Q and height V:

$$W = \tfrac{1}{2} \times \text{base} \times \text{height} \quad \Rightarrow \quad \boxed{W = \tfrac{1}{2}QV}$$

The ½ is there because, on average, each coulomb was pushed on against only half of the final p.d.

Deriving W = ½CV² and W = ½Q²/C

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the result above and the definition of capacitance:

$$W = \tfrac{1}{2}QV, \qquad C = \frac{Q}{V}$$

2. To get rid of Q, write $Q = CV$ and substitute:

$$W = \tfrac{1}{2}(CV)V \quad \Rightarrow \quad \boxed{W = \tfrac{1}{2}CV^2}$$

3. To get rid of V instead, write $V = Q/C$ and substitute:

$$W = \tfrac{1}{2}Q\left(\frac{Q}{C}\right) \quad \Rightarrow \quad \boxed{W = \frac{1}{2}\frac{Q^2}{C}}$$

All three are the same energy. Pick the one that uses the two quantities you already know.

Energy stored

FORMULA SHEET

$$W = \tfrac{1}{2}QV$$

Energy stored

FORMULA SHEET

$$W = \tfrac{1}{2}CV^2$$

Energy stored

FORMULA SHEET

$$W = \frac{1}{2}\frac{Q^2}{C}$$

EXAM TIP

For a given capacitor the energy goes as V^2: double the p.d. and you store **four** times the energy. This is why defibrillator and camera-flash capacitors are charged to hundreds of volts.

COMMON MISTAKE

$W = QV$ is the work done by a battery moving charge Q through a constant p.d. V. For a capacitor the p.d. rises from zero as the charge goes on, so the energy stored is only $\tfrac{1}{2}QV$. Don't drop the half.

WORKED EXAMPLE

A 220 μF capacitor is charged to a p.d. of 9.0 V. Calculate (a) the charge stored and (b) the energy stored. (c) The p.d. is increased to 18 V. How much extra energy is stored?

Show solution

(a) $Q = CV = 220 \times 10^{-6} \times 9.0 = 1.98 \times 10^{-3}\ \text{C} \approx 2.0\ \text{mC}$

(b) $W = \tfrac{1}{2}CV^2 = \tfrac{1}{2} \times 220 \times 10^{-6} \times 9.0^2 = 8.9 \times 10^{-3}\ \text{J}$

(c) Energy goes as V^2, so doubling V gives four times the energy: $4 \times 8.91 \times 10^{-3} = 3.56 \times 10^{-2}\ \text{J}$. The extra energy is $3.56 \times 10^{-2} - 0.89 \times 10^{-2} = 2.7 \times 10^{-2}\ \text{J}$ (three times the original, not one).

Charging and discharging through a resistor

In real circuits a capacitor charges and discharges through a resistor, and the resistor sets how fast it happens. The key to every graph in this topic is to ask: **what is the p.d. across the resistor right now?** That p.d. decides the current, $I = V_R/R$.

Charging

At the instant the switch closes, the capacitor is empty, so $V_C = 0$ and the whole battery e.m.f. is across the resistor. The current starts at its biggest value, $I_0 = V_0/R$. As charge builds up, V_C rises. Kirchhoff's second law says $V_0 = V_C + V_R$, so V_R has to fall, and so does the current. Less current means charge arrives more slowly, so V_C rises more slowly. The capacitor creeps towards V_0 but the closer it gets, the slower it goes.

Discharging

Now switch the battery out and connect the charged capacitor across the resistor alone. The capacitor drives a current round the circuit, in the **opposite** direction to the charging current. The p.d. across the resistor is now the p.d. across the capacitor, so $I = V_C/R = Q/RC$. The current is proportional to the charge that is left. A full capacitor empties quickly; a nearly empty one has hardly any p.d. to push with and empties slowly.

"The rate of loss is proportional to the amount left" is the signature of **exponential decay**. You will meet the same pattern in radioactive decay in Unit 5.

KEY IDEA

The shapes to remember

	Charging	Discharging
Charge Q and p.d. V_C	rise quickly, then level off at $Q_0 = CV_0$ and V_0	fall exponentially towards zero
Current I	starts at V_0/R and falls exponentially to zero	starts at V_0/R in the opposite direction and falls exponentially to zero
p.d. across R	$V_0 - V_C$, falls to zero	equal to V_C

Q and V_C always have the same shape because $V_C = Q/C$. The current is the gradient of the Q–t graph ($I = \Delta Q/\Delta t$), so it is largest where the Q–t graph is steepest.

COMMON MISTAKE

When you sketch the discharge current, check whether the question wants its **magnitude** (drawn above the axis) or its **direction** too (drawn below the axis, because it flows the opposite way to the charging current). Also, changing R does not change the final charge $Q_0 = CV_0$. It only changes how long it takes to get there.

The time constant RC

How fast all of this happens depends on just one number, the product RC. A bigger resistance means a smaller current, so charge moves more slowly. A bigger capacitance means more charge has to move for every volt of change. Either way, it takes longer. RC is called the **time constant**, and it really is a time:

$$[RC] = \Omega \times \mathrm{F} = \frac{\mathrm{V}}{\mathrm{A}} \times \frac{\mathrm{C}}{\mathrm{V}} = \frac{\mathrm{C}}{\mathrm{A}} = \frac{\mathrm{A\ s}}{\mathrm{A}} = \mathrm{s}$$

DEFINITION

The **time constant** RC of a capacitor–resistor circuit is the time taken for the charge (or current, or p.d.) of a discharging capacitor to fall to $1/e$ of its initial value, which is about **37%**. While charging from empty, it is the time for the charge to reach $1 - 1/e$, about **63%**, of its final value.

Time	Fraction left when discharging, $e^{-t/RC}$	Fraction reached when charging
RC	0.368 (37%)	0.632 (63%)
$2RC$	0.135	0.865
$3RC$	0.050	0.950
$5RC$	0.007	0.993

After about $5RC$ a capacitor is, for practical purposes, fully charged or fully discharged. An exponential never actually reaches zero, but after $5RC$ less than 1% is left.

Half-life

An exponential decay loses the **same fraction** in every equal interval of time. So there is a fixed time, the half-life $t_{1/2}$, in which the charge always halves: from 100% to 50%, from 50% to 25%, and so on.

Half-life of a discharging capacitor

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 After one half-life, the charge is half its starting value:

$$\tfrac{1}{2}Q_0 = Q_0\, e^{-t_{1/2}/RC} \quad \Rightarrow \quad \tfrac{1}{2} = e^{-t_{1/2}/RC}$$

2 Take natural logs of both sides. $\ln \tfrac{1}{2} = -\ln 2$:

$$-\ln 2 = -\frac{t_{1/2}}{RC} \quad \Rightarrow \quad \boxed{t_{1/2} = RC \ln 2 \approx 0.69\, RC}$$

The starting charge cancelled out, which is why the half-life is the same however full the capacitor is.

EXAM TIP

Two quick ways to read RC from a discharge graph: find where the curve has fallen to 37% of its starting value, or draw the tangent at $t = 0$ and see where it cuts the time axis. Both give $t = RC$. Using the 37% point is usually more accurate, because drawing a tangent by eye is hard.

The exponential discharge equations

For a capacitor discharging through a resistor, the charge left after time t is given on the formula sheet:

Capacitor discharge

FORMULA SHEET

$$Q = Q_0\, e^{-t/RC}$$

Q_0: charge at $t = 0$ / C t: time since discharge began / s RC: time constant / s

The power $-t/RC$ is "how many time constants have passed", with a minus sign because the charge is falling. Each time constant multiplies what's left by $1/e = 0.368$. The p.d. and current behave in exactly the same way, and you need to be able to show why.

Deriving V = V₀e^(−t/RC) and I = I₀e^(−t/RC)

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the discharge equation and divide both sides by C. Since $V = Q/C$ at every instant:

$$\frac{Q}{C} = \frac{Q_0}{C}\, e^{-t/RC} \quad \Rightarrow \quad \boxed{V = V_0\, e^{-t/RC}}$$

$V_0 = Q_0/C$ is the p.d. at the start. C is a constant, so dividing by it doesn't change the shape.

2. During discharge the capacitor is connected directly across the resistor, so the p.d. across the resistor equals the p.d. across the capacitor. Ohm's law gives the current:

$$I = \frac{V}{R} = \frac{V_0}{R}\, e^{-t/RC}$$

3. V_0/R is the current at the start, I_0, so:

$$\boxed{I = I_0\, e^{-t/RC}}$$

FORMULA SHEET

$$I = I_0\, e^{-t/RC}$$

FORMULA SHEET

$$V = V_0\, e^{-t/RC}$$

Turning the curve into a straight line

Exponential curves are hard to analyse by eye. Taking the natural log of both sides turns them into straight lines, which is the trick behind Core Practical 11.

Deriving the log equations

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Take natural logs of both sides of $Q = Q_0 e^{-t/RC}$:

$$\ln Q = \ln\left(Q_0\, e^{-t/RC}\right)$$

2. The log of a product is the sum of the logs, $\ln(ab) = \ln a + \ln b$:

$$\ln Q = \ln Q_0 + \ln\left(e^{-t/RC}\right)$$

3 $\ln$ undoes e, so $\ln(e^x) = x$:

$$\boxed{\ln Q = \ln Q_0 - \frac{t}{RC}}$$

4 Exactly the same steps on the p.d. and current equations give:

$$\boxed{\ln V = \ln V_0 - \frac{t}{RC}} \qquad \boxed{\ln I = \ln I_0 - \frac{t}{RC}}$$

Compare with $y = c + mx$: plotting $\ln V$ against t gives a straight line with gradient $-1/RC$ and intercept $\ln V_0$.

FORMULA SHEET

$$\ln Q = \ln Q_0 - \frac{t}{RC}$$

FORMULA SHEET

$$\ln I = \ln I_0 - \frac{t}{RC}$$

FORMULA SHEET

$$\ln V = \ln V_0 - \frac{t}{RC}$$

EXAM TIP

To find a **time**, rearrange the log equation: $t = RC\ln(V_0/V)$. Keep V_0/V bigger than 1 and the answer comes out positive. To find a **value** after a time, use the exponential form directly, and work out t/RC first so you don't mistype the power into your calculator.

WORKED EXAMPLE

A 470 μF capacitor is charged to 12.0 V and then discharged through a 2.2 kΩ resistor. Calculate (a) the time constant, (b) the initial current, (c) the p.d. after 2.0 s, (d) the time for the p.d. to fall to 3.0 V and (e) the percentage of the original energy still stored at that moment.

Show solution

(a) $RC = 2200 \times 470 \times 10^{-6} = 1.034 \text{ s} \approx 1.03 \text{ s}$

(b) $I_0 = V_0/R = 12.0/2200 = 5.45 \times 10^{-3} \text{ A} = 5.45 \text{ mA}$

(c) $t/RC = 2.0/1.034 = 1.934$, so $V = 12.0\, e^{-1.934} = 12.0 \times 0.1445 = 1.73 \text{ V}$

(d) $t = RC \ln(V_0/V) = 1.034 \times \ln(12.0/3.0) = 1.034 \times 1.386 = 1.43 \text{ s}$

(e) $W \propto V^2$, so the fraction left is $(3.0/12.0)^2 = 1/16 = 6.25\%$. Three quarters of the p.d. has gone, but nearly 94% of the energy has gone.

WORKED EXAMPLE

A camera flash uses a 150 μF capacitor charged to 300 V. When the flash fires, 90% of the stored energy is released in 1.0 ms. Calculate the mean power delivered to the flash tube, and explain why the camera cannot simply use its 3 V battery directly.

Show solution

Energy stored: $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 150 \times 10^{-6} \times 300^2 = 6.75 \text{ J}$

Energy released: $0.90 \times 6.75 = 6.08 \text{ J}$, so mean power $= 6.08/1.0 \times 10^{-3} = 6.1 \times 10^3 \text{ W}$

The battery has enough energy, but its internal resistance limits the current it can deliver, so it can only supply energy slowly (a few watts). The capacitor is charged slowly from the battery over a few seconds, then delivers the energy very quickly through the low resistance of the flash tube (small RC).

Core Practical 11: capacitor discharge

CORE PRACTICAL

Use an oscilloscope or data logger to display and analyse the p.d. across a capacitor as it charges and discharges through a resistor

Method. Connect a capacitor (for example 470 μF), a resistor (for example 47 kΩ), a battery and a two-way switch so that the switch either connects the capacitor to the battery or connects it across the resistor. Connect a voltage sensor and data logger (or an oscilloscope) across the capacitor. Charge the capacitor fully, then flip the switch to discharge it through the resistor, logging the p.d. at regular intervals. Repeat with the switch flipped back to record the charging curve.

Using an oscilloscope. For a small time constant (for example 0.1 μF and 10 kΩ, RC = 1 ms), replace the battery and switch with a square-wave signal generator. The capacitor charges and discharges every cycle and the oscilloscope shows a steady trace. Read V at a series of times from the time-base.

Analysis. Since $\ln V = \ln V_0 - t/RC$, plot $\ln V$ **against** t. A straight line confirms the decay is exponential. The **gradient is** $-1/RC$, so $RC = -1/\text{gradient}$, and with R known, $C = RC/R$. The intercept is $\ln V_0$. You can also check by reading the time for V to fall to 37% of V_0.

- A data logger removes reaction-time error and can take hundreds of readings. If you use a voltmeter and stopwatch instead, choose a large RC (tens of seconds) and take readings at fixed intervals.
- Stop using readings once V is very small: the resolution of the meter becomes a large fraction of the reading and the $\ln V$ points scatter.
- The voltmeter or logger has a finite resistance in parallel with R. If it is not much bigger than R, the capacitor also discharges through the meter, the measured RC is too small and so is C. This is a systematic error: use a high-resistance logger, or a smaller R.
- Electrolytic capacitors have a large tolerance (often ±10% to ±20%), so the measured C can differ from the labelled value by this much.
- **Safety:** connect electrolytic capacitors the right way round and never above their rated voltage, as they can overheat and burst. Discharge large capacitors before handling them.

WORKED EXAMPLE

In Core Practical 11 a student discharges a capacitor through a 47 kΩ resistor and plots $\ln(V/\text{V})$ against t. The line passes through (0 s, 1.79) and (60 s, 0.29). Determine the capacitance, and the p.d. the capacitor was charged to.

Show solution

Gradient $= \dfrac{0.29 - 1.79}{60 - 0} = -0.0250\ \text{s}^{-1}$

Gradient $= -1/RC$, so $RC = 1/0.0250 = 40\ \text{s}$, and $C = \dfrac{40}{47 \times 10^3} = 8.5 \times 10^{-4}\ \text{F} = 850\ \mu\text{F}$

Intercept $= \ln V_0 = 1.79$, so $V_0 = e^{1.79} = 6.0\ \text{V}$

Quick checks

QUICK CHECK

A charged capacitor discharges through a resistor. What fraction of its original charge is left after a time 2RC?

- 0.25
- 0.37
- 0.14
- 0.63

$Q/Q_0 = e^{-2} = 0.135$. Each time constant multiplies the charge by 0.37, so two of them give $0.37^2 \approx 0.14$. 0.25 would be two **half-lives**, and a half-life is shorter than RC.

QUICK CHECK

The p.d. across a capacitor is doubled. What happens to the energy stored?

- It stays the same
- It doubles
- It increases by a factor of 4
- It halves

$W = \frac{1}{2}CV^2$ and C is fixed, so $W \propto V^2$. Doubling V also doubles Q, and $W = \frac{1}{2}QV$ gives the same factor of 4.

QUICK CHECK

Which of these is equivalent to the unit of the time constant, Ω × F?

- A s^{-1}
- s
- C V
- J s^{-1}

$\Omega = \mathrm{V\ A^{-1}}$ and $\mathrm{F = C\ V^{-1}}$, so $\Omega\,\mathrm{F = C\ A^{-1} = (A\ s)\ A^{-1} = s}$.

QUICK CHECK

An uncharged capacitor is connected in series with a resistor and a battery. At the instant the switch is closed, which statement is correct?

- The current is zero because the capacitor has no charge.
- The current is at its maximum, V_0/R.
- The p.d. across the capacitor is V_0.
- The p.d. across the resistor is zero.

With no charge on it, the capacitor has no p.d. across it, so the whole e.m.f. is across the resistor. The current is V_0/R, its largest value. It falls as V_C builds up.

QUICK CHECK

A student plots ln I against t for a capacitor discharging through a resistor. What is the gradient of the line?

- −RC
- ln I_0
- −1/RC
- 1/RC

$\ln I = \ln I_0 - t/RC$ has the form $y = c + mx$ with $m = -1/RC$. The intercept is $\ln I_0$.

QUICK CHECK

A capacitor is charged to a fixed p.d. and discharged through a resistor. The resistor is replaced with one of half the resistance and the experiment is repeated. Which statement is correct?

- The initial current halves and the total charge that flows doubles.
- The time constant doubles.
- The initial current doubles and the total charge that flows is unchanged.
- The energy dissipated in the resistor halves.

$I_0 = V_0/R$ doubles and RC halves, so the discharge is twice as fast. But the capacitor started with the same $Q_0 = CV_0$ and energy $\frac{1}{2}CV_0^2$, and all of it still flows through the resistor (the area under the I–t graph is unchanged).